

```
In [3]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [5]: file_path = r"C:\Users\komal\Downloads\Unemployment in India.csv"
df = pd.read_csv(r"C:\Users\komal\Downloads\Unemployment in India.csv")
```

```
In [7]: print("Column Names:", df.columns)
```

```
Column Names: Index(['Region', ' Date', ' Frequency', ' Estimated Unemployment Ra
te (%)',
                    ' Estimated Employed', ' Estimated Labour Participation Rate (%)',
                    'Area'],
                  dtype='object')
```

```
In [9]: df.columns = df.columns.str.strip()
```

```
In [13]: print("\nFirst 5 rows of the dataset:")
print(df.head())
```

First 5 rows of the dataset:

	Region	Date	Frequency	Estimated Unemployment Rate (%) \
0	Andhra Pradesh	31-05-2019	Monthly	3.65
1	Andhra Pradesh	30-06-2019	Monthly	3.05
2	Andhra Pradesh	31-07-2019	Monthly	3.75
3	Andhra Pradesh	31-08-2019	Monthly	3.32
4	Andhra Pradesh	30-09-2019	Monthly	5.17

	Estimated Employed	Estimated Labour Participation Rate (%)	Area
0	11999139.0	43.24	Rural
1	11755881.0	42.05	Rural
2	12086707.0	43.50	Rural
3	12285693.0	43.97	Rural
4	12256762.0	44.68	Rural

```
In [15]: print("\nMissing values in each column:")
print(df.isnull().sum())
```

Missing values in each column:

Region	14
Date	14
Frequency	14
Estimated Unemployment Rate (%)	14
Estimated Employed	14
Estimated Labour Participation Rate (%)	14
Area	14
dtype:	int64

```
In [17]: df = df.dropna()
```

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In [19]: df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
```

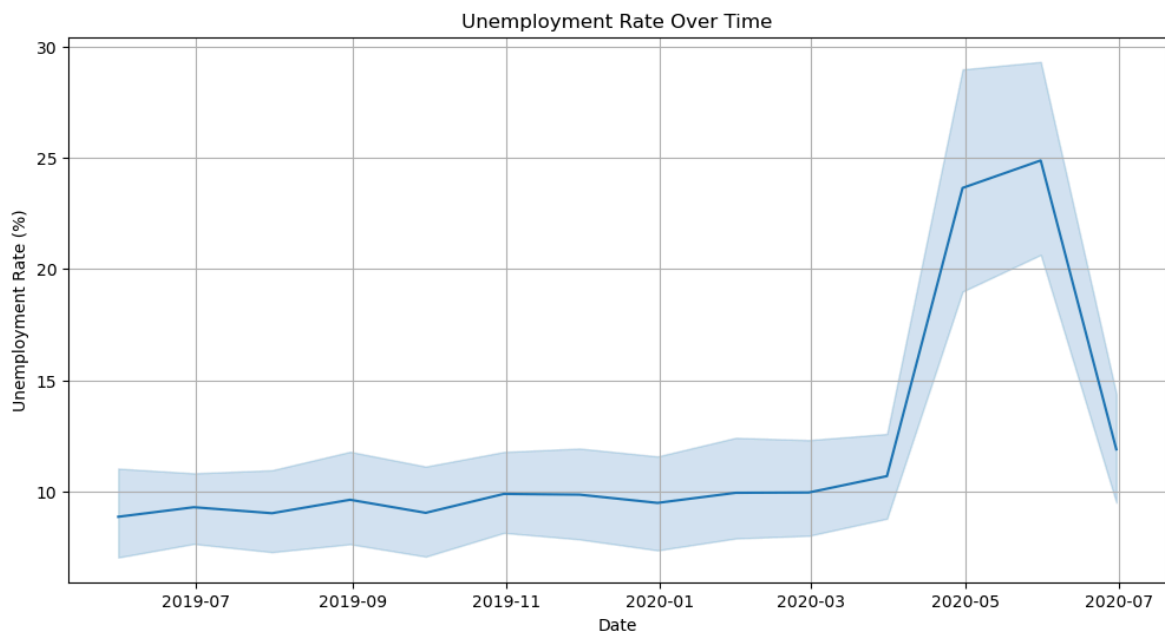
```
In [21]: print("\nBasic Statistics:")
print(df.describe())
```

Basic Statistics:

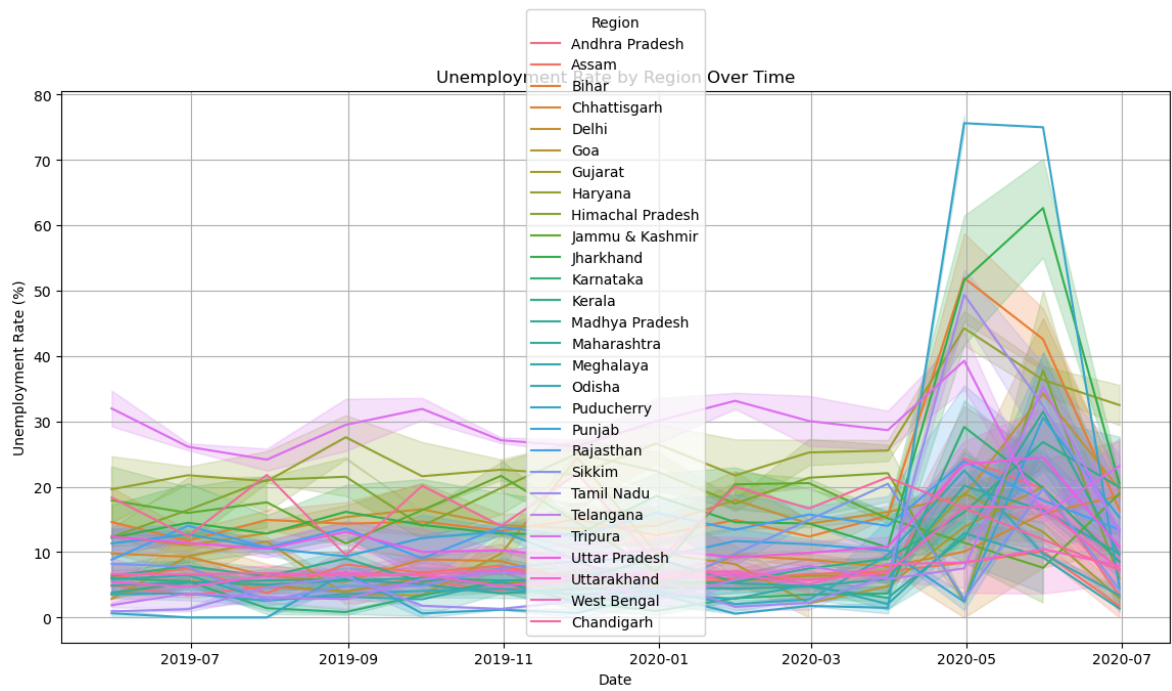
	Date	Estimated Unemployment Rate (%) \
count	740	740.000000
mean	2019-12-12 18:36:58.378378496	11.787946
min	2019-05-31 00:00:00	0.000000
25%	2019-08-31 00:00:00	4.657500
50%	2019-11-30 00:00:00	8.350000
75%	2020-03-31 00:00:00	15.887500
max	2020-06-30 00:00:00	76.740000
std	NaN	10.721298

	Estimated Employed	Estimated Labour Participation Rate (%)
count	7.400000e+02	740.000000
mean	7.204460e+06	42.630122
min	4.942000e+04	13.330000
25%	1.190404e+06	38.062500
50%	4.744178e+06	41.160000
75%	1.127549e+07	45.505000
max	4.577751e+07	72.570000
std	8.087988e+06	8.111094

```
In [23]: plt.figure(figsize=(12, 6))
sns.lineplot(x='Date', y='Estimated Unemployment Rate (%)', data=df)
plt.title('Unemployment Rate Over Time')
plt.xlabel('Date')
plt.ylabel('Unemployment Rate (%)')
plt.grid(True)
plt.show()
```



```
In [30]: if 'Region' in df.columns:
plt.figure(figsize=(14, 7))
sns.lineplot(x='Date', y='Estimated Unemployment Rate (%)', hue='Region', data=df)
plt.title('Unemployment Rate by Region Over Time')
plt.xlabel('Date')
plt.ylabel('Unemployment Rate (%)')
plt.legend(title='Region')
plt.grid(True)
plt.show()
```



In []: